77 Material Data Sheet

A MnZn ferrite for use in a wide range of high and low flux density inductive designs for frequencies up to 100 kHz.

Available in 77 material:
Pot Cores
E&I Cores
U Cores
Rods
Toroids
Bobbins

77 Material Characteristics

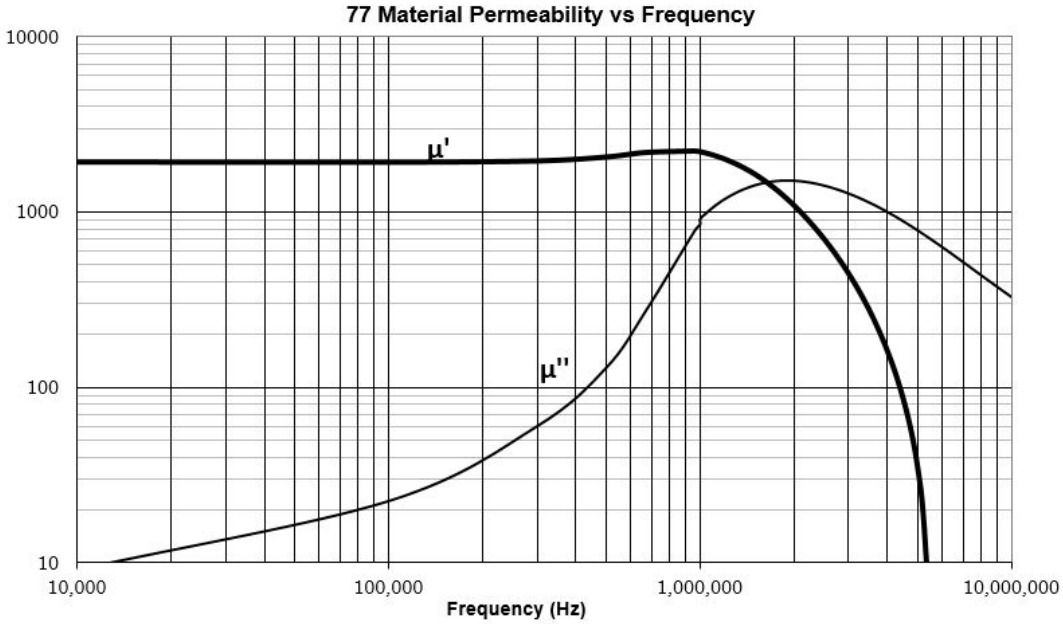
Property	Unit	Symbol	Value
Initial Permeability @ B < 10 gauss		μ_i	2000
Flux Density @ Field Strength	Gauss Oersted	B H	5100 5
Residual Flux Density	Gauss	B_r	1800
Coercive Force	Oersted	H_c	0.25
Loss Factor @ Frequency	10^{-6} MHz	Tan δ/ μ_i	15 0.1
Temperature Coefficient of Initial Permeability (20 -70°C)	%/°C		1.2
Curie Temperature	°C	T_c	>200
Resistivity	ohm-cm	ρ	100

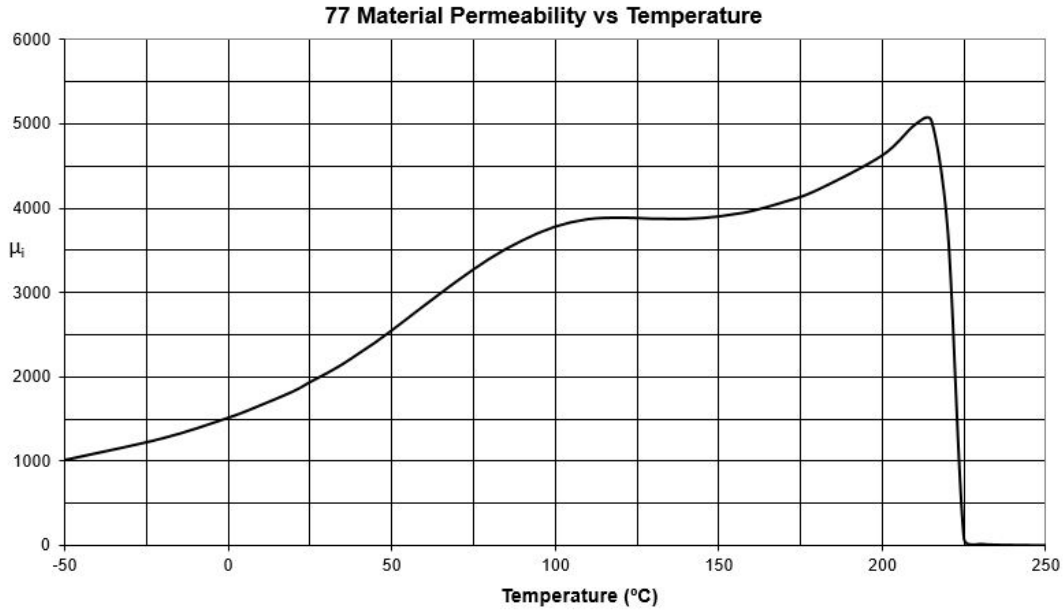
**** Characteristic curves are measured on standard Toroids (18/10/6 mm) at 25°C and 10 kHz unless otherwise indicated. Impedance characteristics are measured on standard shield beads (3.5/1.3/6.0 mm)

unless otherwise indicated.

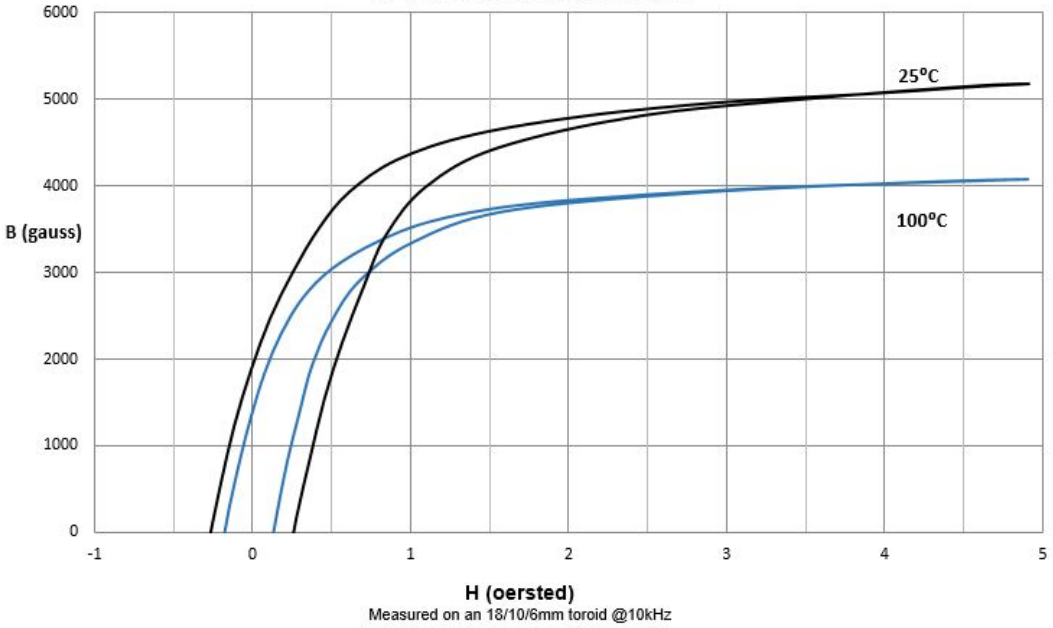
Material Safety Data Sheet (MSDS)

Click here to download Complex Permeability vs. Frequency (CSV)



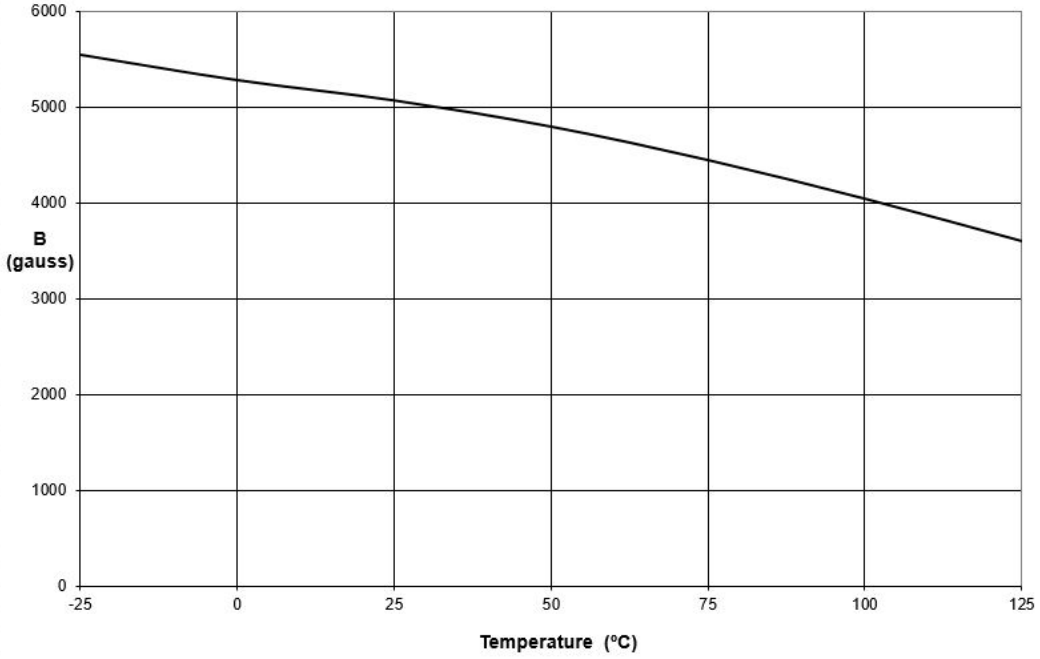


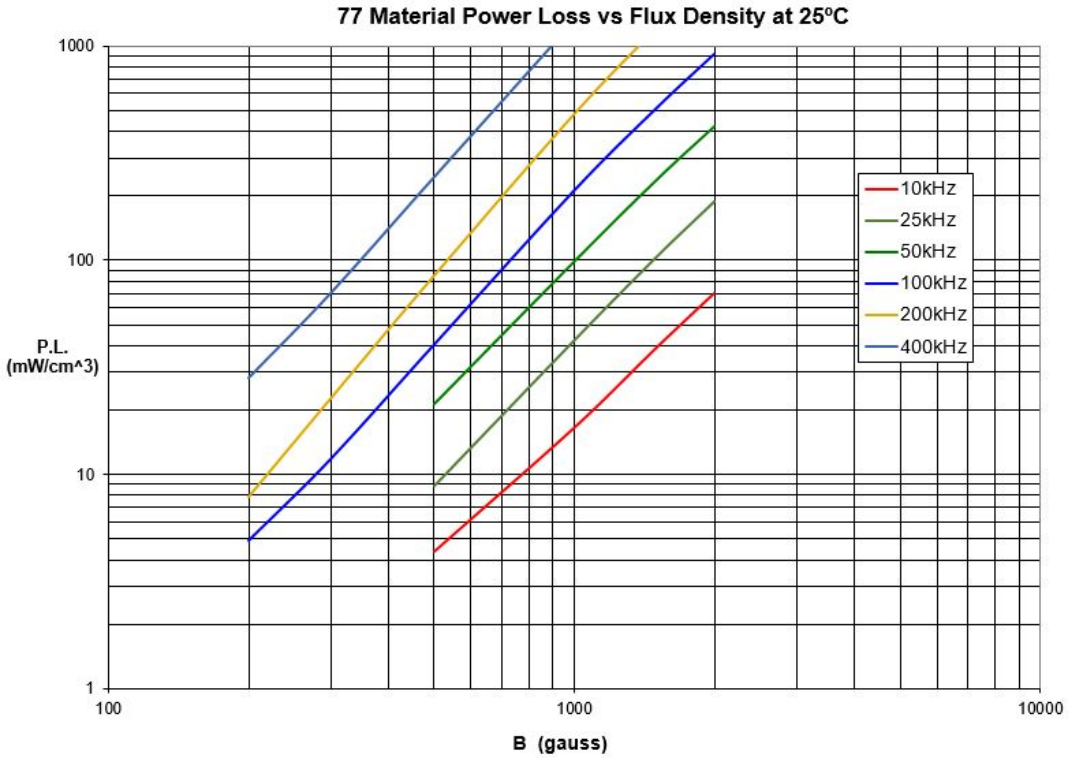
77 Material Hysteresis Loop

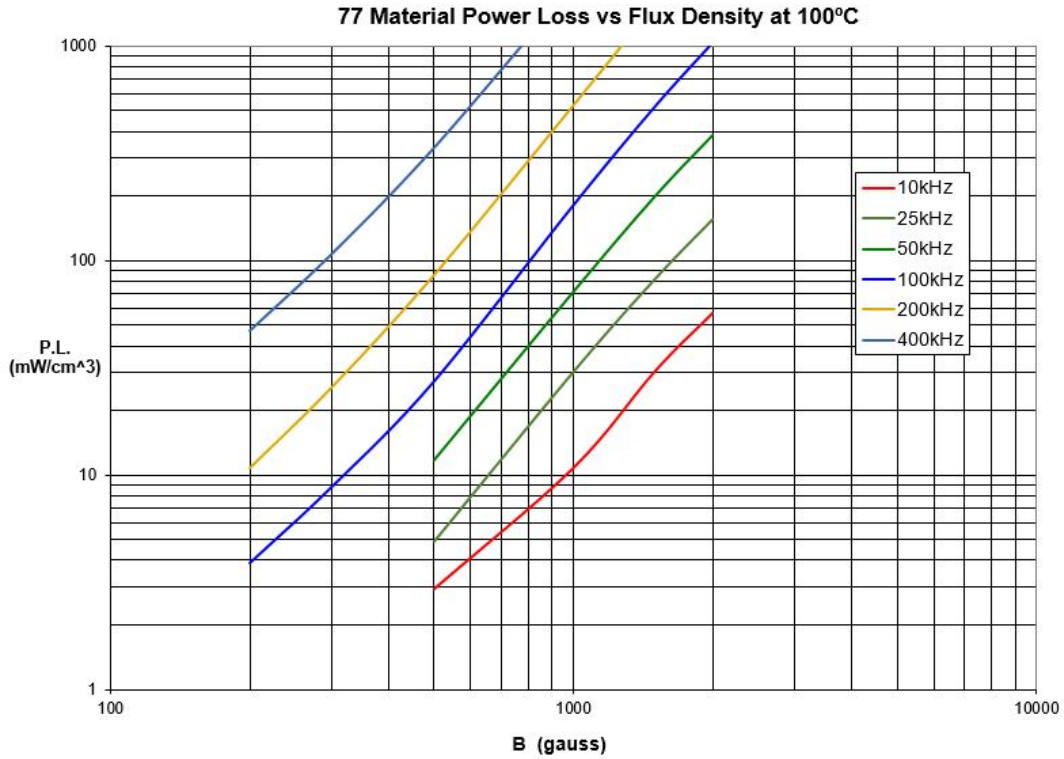


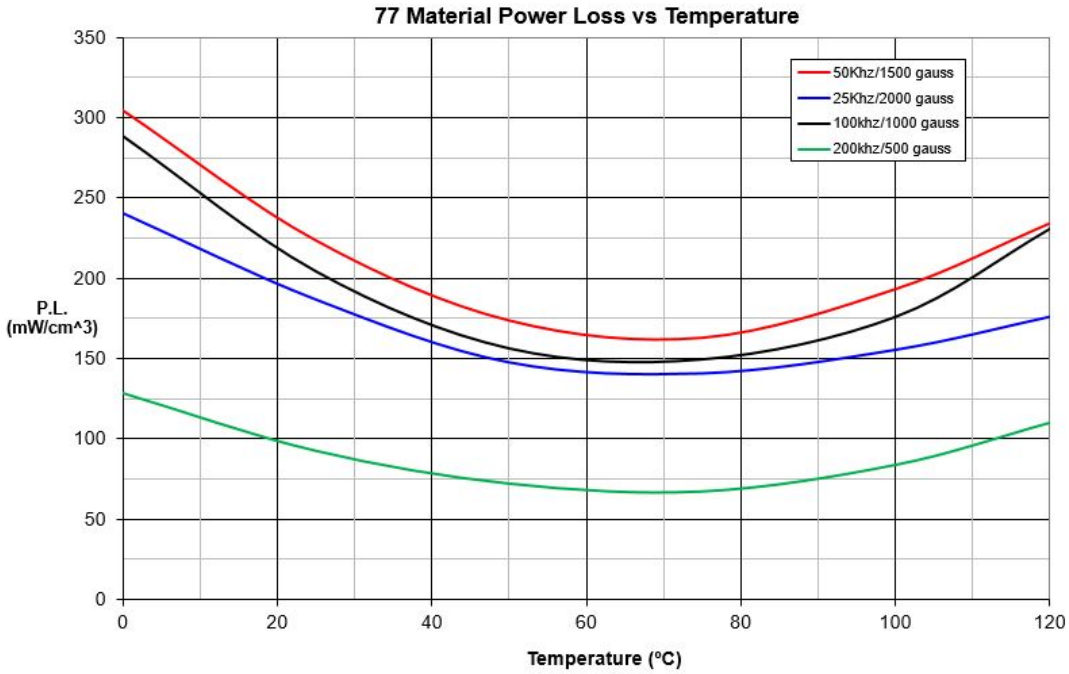
Measured on an 18/10/6mm toroid @10kHz

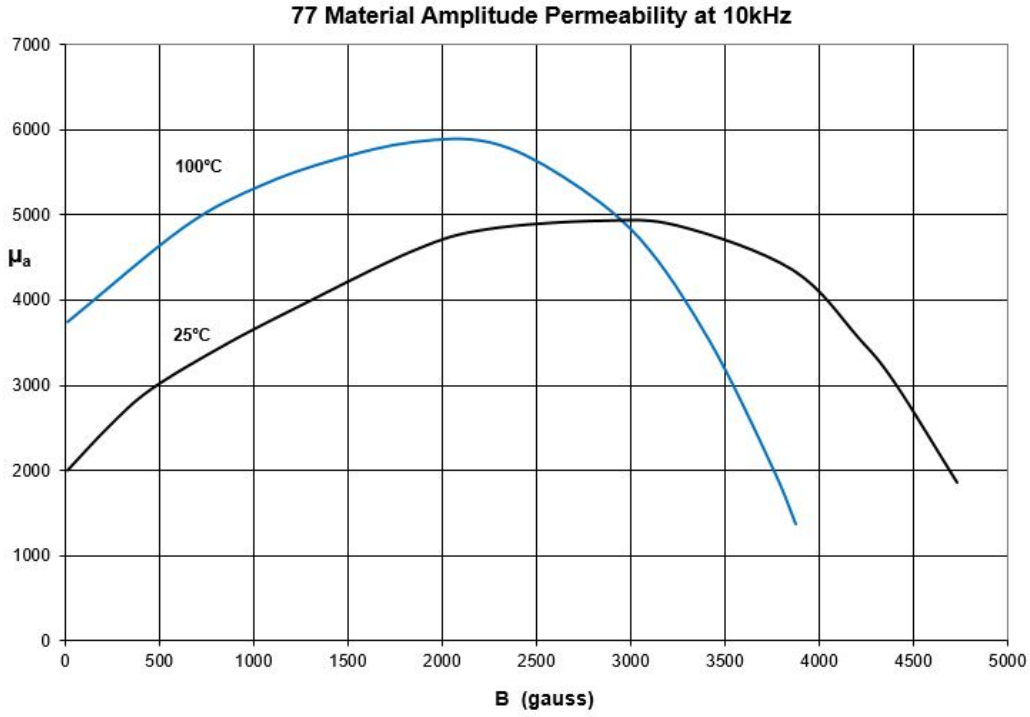
77 Material Flux Density vs Temperature



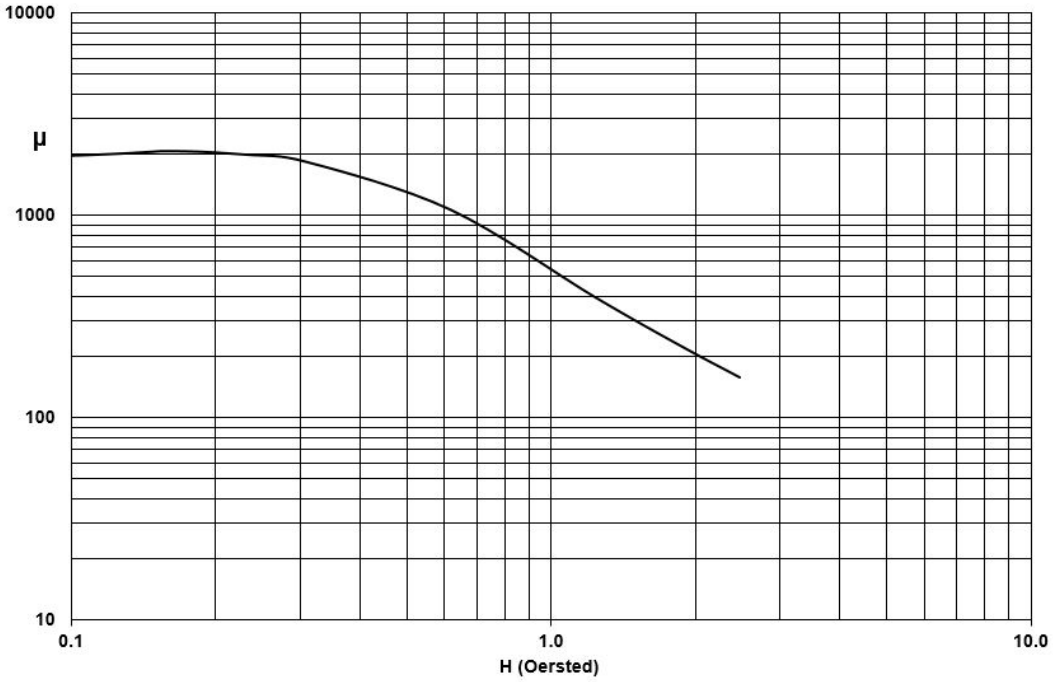

77 Material Power Loss vs Flux Density at 25ºC
1000
100
10
1
P.L.
(mW/cm^3)
100
1000
10000
B (gauss)
10kHz
25kHz
50kHz
100kHz
200kHz
400kHz


77 Material Power Loss vs Flux Density at 100ºC
1000
100
10
1
P.L.
(mW/cm^3)
100
1000
10000
B (gauss)
10kHz
25kHz
50kHz
100kHz
200kHz
400kHz


77 Material Power Loss vs Temperature
50Khz/1500 gauss
25Khz/2000 gauss
100khz/1000 gauss
200khz/500 gauss
P.L.
(mW/cm^3)
350
300
250
200
150
100
50
0
0
20
40
60
80
100
120
Temperature (°C)


77 Material Amplitude Permeability at 10kHz
100°C
25°C
μa
7000
6000
5000
4000
3000
2000
1000
0
0
500
1000
1500
2000
2500
3000
3500
4000
4500
5000
B (gauss)

77 Material Incremental Permeability vs Field Strength



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